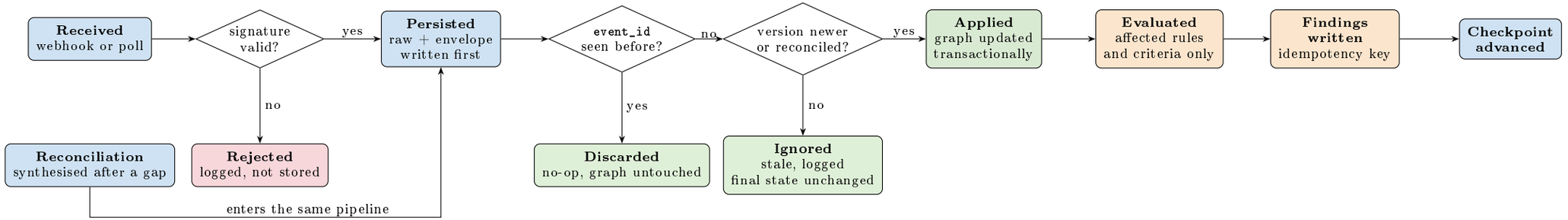
(a) Event processing — one delivery, from receipt to checkpoint (sec. 8.4)



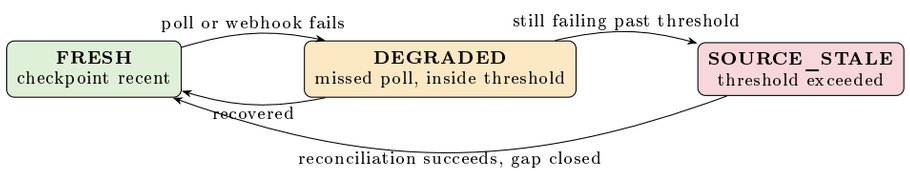
Step order is the guarantee. The envelope is persisted *before* evaluation, so a crash between steps cannot lose an event that was already accepted. The checkpoint advances *last*, so a crash before it causes re-processing, which is safe because the duplicate branch is a no-op.

Three different "nothing happened" outcomes. Rejected – never trusted. **Discarded** – already applied once. **Ignored** – superseded by a newer version. Each is counted separately in the reliability suite, because collapsing them hides real failures.

Measured here (sec. 12.1).
0 duplicate findings from duplicate delivery
100 % final-state convergence under reordering
$\geq$ 99 % incremental vs. full-rescan equivalence
ingestion p50 / p95 latency, vendor delay separated from internal delay via `occurred_at` vs `received_at`

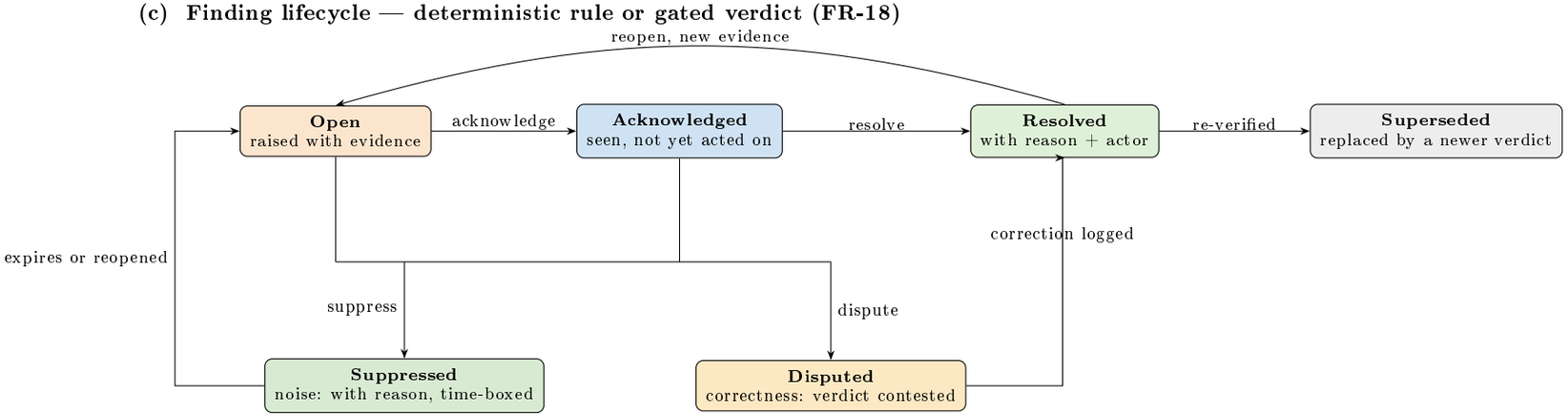
Reconciliation is not a retry. A retry resends something that failed. Reconciliation polls the vendor, compares against the stored checkpoint and *manufactures* the event the vendor never sent, so it carries `action = reconciled` and is applied even when its timestamp is older than the stored version.

(b) Source freshness, per connected resource (FR-16)



What SOURCE_STALE does downstream. Every artifact whose source is stale is marked, and any finding that would be computed from it is downgraded to **SOURCE_STALE** rather than served from old data (sec. 11.7). The reviewer is told the data cannot establish current state — the system never silently presents a stale verdict as current.

(c) Finding lifecycle — deterministic rule or gated verdict (FR-18)



Why suppress and dispute do not connect. They answer different questions. *Suppressed* means the finding is correct but unwanted here — a known parallel refactor, an accepted deviation — and it is a noise control, time-boxed, returning to **Open** when it expires. *Disputed* means the verdict itself is wrong. A finding can be either, from **Open** or from **Acknowledged**, and neither is a precondition for the other. Suppressing a finding in order to dispute it would discard the correction that dispute exists to capture.

Disputed is a data path, not just a status (FR-20). A contested verdict is logged with the reviewer's correction and becomes a labelled example for the next adaptation round, under the same repository-level split discipline (sec. 28.3). This is the only route by which production use feeds the training set.

Superseding may occur from any non-terminal state when re-verification on a newer commit produces a replacement; the arrow is drawn from **Resolved** as the common case, and the replacement is recorded by `findings.superseded_by_id` rather than by overwriting the original.

Every transition appends to `audit_log` with actor, reason and an evidence snapshot, so a finding can always be re-read against the artifact versions it cited.